

$$1) P_A + \rho_n g h + \rho_w g (10 \times 10^{-3}) - \rho_k g (36 \times 10^{-3}) - \rho_n g (10) = P_B$$

$$P_B - P_A = 0.6 (9.81) (36 - 10) \times 10^{-3} + 1000 (9.81) (10 \times 10^{-3}) \\ - 800 (9.81) (36 \times 10^{-3}) - 0.6 (9.81) (10)$$

$$= -243.134964$$

$$\approx -243.13 \text{ Pa}$$

$$\Delta P = \rho_n g (-10 + h_L)$$

$$\Delta P = \rho_n g \left(f \frac{L}{D} \frac{v^2}{2g} - 10 \right)$$

$$243.13 = 0.6 (9.81) \left(0.016 \left(\frac{30}{10 \times 10^{-2}} \right) \left(\frac{v^2}{2 \times 9.81} \right) - 10 \right)$$

$$243.13 = 1.44 v^2 - 58.86$$

$$v = 11.31232624 \approx 11.31 \text{ m s}^{-1}$$

$$\dot{m} = \rho_n Q = \rho_n v A$$

$$= 0.6 (11.31) \left(\pi \left(\frac{10 \times 10^{-2}}{2} \right)^2 \right)$$

$$= 0.053330808152$$

$$\approx 0.053 \text{ kg s}^{-1}$$

$$\begin{aligned}
 2) Q &= 30 \text{ L s}^{-1} \\
 &= 30 \times 10^{-3} \text{ m}^3 \text{ s}^{-1} \\
 &= 0.03 \text{ m}^3 \text{ s}^{-1}
 \end{aligned}$$

$$\begin{aligned}
 V_1 &= \frac{Q}{A} = \frac{0.03}{\pi \left(\frac{100 \times 10^{-3}}{2} \right)^2} \\
 &= 3.819718634 \text{ m s}^{-1} \\
 &\approx 3.82 \text{ m s}^{-1}
 \end{aligned}$$

$$\begin{aligned}
 V_2 &= \frac{Q}{A} = \frac{0.03}{\pi \left(\frac{75 \times 10^{-3}}{2} \right)^2} \\
 &= 6.790610905 \text{ m s}^{-1} \\
 &\approx 6.79 \text{ m s}^{-1}
 \end{aligned}$$

$$\begin{aligned}
 Re_1 &= \frac{\rho V D}{\mu} \\
 &= \frac{1000 \times 3.82 \times 0.1}{1 \times 10^{-3}} \\
 &= 381971.8634 \\
 &\approx 381972 > 4000
 \end{aligned}$$

$$\begin{aligned}
 Re_2 &= \frac{\rho V D}{\mu} \\
 &= \frac{1000 \times 6.79 \times 0.1}{1 \times 10^{-3}} \\
 &= 679061.0905 \\
 &\approx 679061 > 4000
 \end{aligned}$$

Major losses:

$$\begin{aligned}
 h_{L1} &= f \frac{L}{D} \frac{V^2}{2g} \\
 &= \frac{0.316}{Re^{\frac{1}{4}}} \frac{L}{D} \frac{V^2}{2g} \\
 &= \frac{0.316}{381972^{\frac{1}{4}}} \left(\frac{7}{0.1} \right) \left(\frac{3.82^2}{2 \times 9.81} \right) \\
 &= 0.6616693581 \\
 &\approx 0.662 \text{ m}
 \end{aligned}$$

$$\begin{aligned}
 h_{L1} &= f \frac{L}{D} \frac{V^2}{2g} \\
 &= \frac{0.316}{Re^{\frac{1}{4}}} \frac{L}{D} \frac{V^2}{2g} \\
 &= \frac{0.316}{679061^{\frac{1}{4}}} \left(\frac{30}{0.075} \right) \left(\frac{6.79^2}{2 \times 9.81} \right) \\
 &= 10.34876565 \\
 &\approx 10.349 \text{ m}
 \end{aligned}$$

2) Assuming the pressure at the well is 0, energy equation from the well to the pump inlet:

$$0 = \frac{P_1}{\rho g} + \frac{V_1^2}{2g} + z_1 + h_L$$

$$\frac{P_1}{1000 \times 9.81} = - \left(\frac{3.82^2}{2 \times 9.81} + 3 + 0.662 \right)$$

$$P_1 = -43216.10163$$

$$\approx -43.2 \text{ kPa}$$

Energy equation from the delivery flange to the outlet:

$$\frac{P_2}{\rho g} + \frac{V_2^2}{2g} + z_2 - h_{L2} = \frac{P_3}{\rho g} + \frac{V_3^2}{2g} + z_3$$

$$P_2 = \rho g (z_3 - z_2 + h_{L2})$$

$$= 1000(9.81)(15 + 10.349)$$

$$= 248671.391 \text{ Pa}$$

$$\approx 249 \text{ kPa}$$

$$3) Q = 49 \text{ L s}^{-1} = 0.049 \text{ m}^3 \text{ s}^{-1}$$

$$V = \frac{Q}{A} = \frac{0.049}{\pi \left(\frac{0.2}{2}\right)^2} = 1.559718442 \\ \approx 1.56 \text{ m s}^{-1}$$

Energy equation from the upstream end to the discharge end:

$$\frac{P_1}{\rho g} + \frac{V^2}{2g} + z_1 - h_L = \frac{P_2}{\rho g} + \frac{V_2^2}{2g} + z_2$$

$$\frac{P_1}{\rho g} = f \frac{L}{D} \frac{V^2}{2g} - z_1$$

$$40 \times 10^3 = 1000 \times 9.81 f \left(\frac{900}{0.2} \right) \left(\frac{1.56^2}{2 \times 9.81} \right) \\ - 1000 \times 9.81 (10)$$

$$f = 0.02523008687 \\ \approx 0.0252$$

After tapping the water:

$$V_1 = \frac{0.049 + 0.018}{\pi \left(\frac{0.2}{2}\right)^2} = 2.132676237 \\ \approx 2.13 \text{ m s}^{-1}$$

$$\frac{P_1}{\rho g} + \frac{V_1^2}{2g} + z_1 - h_{L1} - h_{L2} = \frac{P_2}{\rho g} + \frac{V_2^2}{2g} + z_2$$

$$\frac{P_1}{\rho g} = \frac{V_2^2 - V_1^2}{2} - z_1 + f \frac{L}{D} \frac{V_1^2}{2g} + f \frac{L}{D} \frac{V_2^2}{2g}$$

$$= \frac{V_2^2 - V_1^2}{2} + f \frac{L}{2D} (V_1^2 + V_2^2) - z_1$$

$$= \rho \left(\frac{1}{2} (V_2^2 - V_1^2) + f \frac{L}{2D} (V_1^2 + V_2^2) - z_1 g \right)$$

$$3) P_1 = \rho \left(\frac{1}{2} (v_2^2 - v_1^2) + f \frac{L}{2D} (v_1^2 + v_2^2) - z, g \right)$$

$$= 1000 \left(\frac{1}{2} (1.56^2 - 2.13^2) + 0.0252 \left(\frac{900 \div 2}{2 \times 0.2} \right) (2.13^2 + 1.56^2) - 10(9.81) \right)$$

$$= 98990.68664 \text{ Pa}$$

$$\approx 98.99 \text{ kPa}$$

$$4) \frac{p_1}{\rho g} + \frac{v_1^2}{2g} + z_1 - h_L = \frac{p_2}{\rho g} + \frac{v_2^2}{2g} + z_2$$

$$-h_L = -9$$

$$h_L = 9$$

Major losses:

$$h_{\text{major}} = f \frac{L_1}{D_1} \frac{v_1^2}{2g} + f \frac{L_2}{D_2} \frac{v_2^2}{2g}$$

$$= 0.01 \left(\frac{15}{0.2} \right) \left(\frac{v_1^2}{2 \times 9.81} \right) + 0.01 \left(\frac{45}{0.25} \right) \left(\frac{v_2^2}{2 \times 9.81} \right)$$

$$= \frac{25}{654} v_1^2 + \frac{10}{109} v_2^2$$

Minor losses:

$$h_{\text{minor}} = \frac{1}{2} \frac{v_2^2}{2g} + \frac{v_2^2}{2g} + \left(1 - \frac{A_1}{A_2} \right)^2 \frac{v_1^2}{2g}$$

$$= \frac{1}{2g} \left(v_2^2 + \left[\frac{1}{2} + \left(1 - \frac{\pi D_1^2}{4 \pi D_2^2} \right)^2 \right] v_1^2 \right)$$

$$= \frac{1}{2g} \left(v_2^2 + \left(\frac{1}{2} + \left(1 - \frac{D_1^2}{D_2^2} \right)^2 \right) v_1^2 \right)$$

$$= \frac{1}{2 \times 9.81} \left(v_2^2 + \left(\frac{1}{2} + \left(1 - \frac{0.2^2}{0.25^2} \right)^2 \right) v_1^2 \right)$$

$$= \frac{50}{981} v_2^2 + \frac{787}{24525} v_1^2$$

$$h_L = h_{\text{minor}} + h_{\text{major}}$$

$$= \frac{50}{981} v_2^2 + \frac{787}{24525} v_1^2 + \frac{25}{654} v_1^2 + \frac{10}{109} v_2^2$$

$$= \frac{140}{981} v_2^2 + \frac{3449}{49050} v_1^2$$

$$4) h_L = \frac{140}{981} v_2^2 + \frac{3449}{49050} v_1^2$$

$$v = \frac{Q}{A} = \frac{Q}{\frac{\pi D^2}{4}} = \frac{4Q}{\pi D^2}$$

$$q = \frac{140}{981} \left(\frac{4Q}{\pi D_2^2} \right)^2 + \frac{3449}{49050} \left(\frac{4Q}{\pi D_1^2} \right)^2$$

$$q = \frac{140}{981} \left(\frac{16Q^2}{\pi^2 (0.25)^4} \right) + \frac{3449}{49050} \left(\frac{16Q^2}{\pi^2 (0.2)^4} \right)$$

$$q = 130.4719389Q^2$$

$$Q = 0.2626411048$$

$$\approx 0.263 \text{ m}^3 \text{ s}^{-1}$$

5) Energy equation from 1 to 2:

$$\frac{P_1}{\rho g} + \frac{V_1^2}{2g} + z_1 - h_{L12} = \frac{P_2}{\rho g} + \frac{V_2^2}{2g} + z_2$$

$$\begin{aligned} h_{L12} &= z_1 - z_2 \\ &= 60 - 0 \\ &= 60 \end{aligned}$$

$$h_{L12} = f_1 \frac{L_1}{D_1} \frac{V_1^2}{2g} + f_2 \frac{L_2}{D_2} \frac{V_2^2}{2g}$$

$$60 = 0.015 \left(\frac{200}{0.1} \right) \frac{V_1^2}{2 \times 9.81} + 0.020 \left(\frac{400}{0.08} \right) \frac{V_2^2}{2 \times 9.81}$$

$$60 = \frac{500}{327} V_1^2 + \frac{5000}{981} V_2^2 \quad - (1)$$

Energy equation from 1 to 3:

$$\frac{P_1}{\rho g} + \frac{V_1^2}{2g} + z_1 - h_{L13} = \frac{P_3}{\rho g} + \frac{V_3^2}{2g} + z_3$$

$$\begin{aligned} h_{L13} &= z_1 - z_3 \\ &= 60 - 20 \\ &= 40 \end{aligned}$$

$$h_{L13} = f_1 \frac{L_1}{D_1} \frac{V_1^2}{2g} + f_3 \frac{L_3}{D_3} \frac{V_3^2}{2g}$$

$$40 = \frac{500}{327} V_1^2 + 0.020 \left(\frac{200}{0.08} \right) \left(\frac{V_3^2}{2 \times 9.81} \right)$$

$$40 = \frac{500}{327} V_1^2 + \frac{2500}{981} V_3^2 \quad - (2)$$

$$5) A_1 V_1 = A_2 V_2 + A_3 V_3$$

$$\frac{\pi D_1^2}{4} V_1 = \frac{\pi D_2^2}{4} V_2 + \frac{\pi D_3^2}{4} V_3$$

$$0.10^2 V_1 = 0.08^2 V_2 + 0.08^2 V_3$$

$$0.01 V_1 = 0.0064 V_2 + 0.0064 V_3$$

$$V_1 = \frac{16}{25} (V_2 + V_3) \quad - (3)$$

$$(1) - (2):$$

$$20 = \frac{5000}{981} V_2^2 - \frac{2500}{981} V_3^2$$

$$\frac{981}{125} = 2V_2^2 - V_3^2$$

$$V_3^2 = 2V_2^2 - \frac{981}{125}$$

$$V_3 = \sqrt{2V_2^2 - \frac{981}{125}} \quad - (4)$$

Sub (4) into (3):

$$V_1 = \frac{16}{25} \left(V_2 + \sqrt{2V_2^2 - \frac{981}{125}} \right) \quad - (5)$$

Sub (5) into (1):

$$60 = \frac{500}{327} \left(\frac{16}{25} \left(V_2 + \sqrt{2V_2^2 - \frac{981}{125}} \right) \right)^2 + \frac{5000}{981} V_2^2$$

$$60 = \frac{1024}{1635} \left(V_2 + \sqrt{2V_2^2 - \frac{981}{125}} \right)^2 + \frac{5000}{981} V_2^2$$

Solving:

$$V_2 = 2.808257625 \approx 2.808 \text{ ms}^{-1}$$

5) From (5):

$$\begin{aligned}v_1 &= \frac{16}{25} \left(v_2 + \sqrt{2v_2^2 - \frac{981}{125}} \right) \\&= \frac{16}{25} \left(2.808 + \sqrt{2(2.808)^2 - \frac{981}{125}} \right) \\&= 3.598929984 \\&\approx 3.599 \text{ m s}^{-1}\end{aligned}$$

From (4):

$$\begin{aligned}v_3 &= \sqrt{2(2.808)^2 - \frac{981}{125}} \\&= 2.815070475 \\&\approx 2.815 \text{ m s}^{-1}\end{aligned}$$

$$\begin{aligned}Q_1 = A_1 v_1 &= \frac{\pi D_1^2}{4} v_1 = \frac{\pi (0.1)^2}{4} (3.599) \\&= 0.02826592999 \\&\approx 0.0283 \text{ m}^3 \text{ s}^{-1}\end{aligned}$$

$$\begin{aligned}Q_2 = A_2 v_2 &= \frac{\pi D_2^2}{4} v_2 = \frac{\pi (0.08)^2}{4} (2.808) \\&= 0.01411584244 \\&\approx 0.0141 \text{ m}^3 \text{ s}^{-1}\end{aligned}$$

$$\begin{aligned}Q_3 = A_3 v_3 &= \frac{\pi D_3^2}{4} v_3 = \frac{\pi (0.08)^2}{4} (2.815) \\&= 0.01415008756 \\&\approx 0.0142 \text{ m}^3 \text{ s}^{-1}\end{aligned}$$